

## Research paper

# Composition optimization of a hypergolic green propellant based on monoethanolamine, n-butanol and 90% hydrogen peroxide

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## ABSTRACT

The design of satellite attitude-control thrusters depends on a trade-off between minimum impulse bit and specific impulse, where the width of pulse maneuvers relies on the combination of delays in the hydraulic system (feed tubes and valves) and the ignition delay time of the propellant used. The most well-established propellants in this context are hydrazine derivatives and nitrogen tetroxide. However, their high toxicity makes satellite integration costly and environmentally hazardous. To replace these propellants, research is focused on developing new hypergolic green propellants, most of which use high-concentration hydrogen peroxide as an oxidizer. In this study, the hypergolic reaction between a blend of n-butanol and monoethanolamine and hydrogen peroxide was catalyzed using copper nitrate trihydrate. The central composite design method was applied to optimize fuel composition using 90% hydrogen peroxide as the oxidizer. The optimization yielded two key outcomes: for ignition delay time (31.5% n-butanol, 60% monoethanolamine, and 8.5% copper nitrate, resulting in an ignition delay time of 21.5 ms with a standard deviation of  $\pm 1.30$  ms and a systematic error of  $\pm 0.4$ ), and for theoretical specific impulse (36% n-butanol, 60% monoethanolamine, and 4% copper nitrate, with an ignition delay time of  $26 \pm 0.4$  ms). For the ignition delay time optimization, an oxidizer-fuel ratio of 4 was selected using CEA NASA software to achieve a maximum theoretical specific impulse of 170.64 s, while for specific impulse optimization, a ratio of 4.4 was chosen, resulting in a specific impulse of 171.58 s. Although the maximum theoretical specific impulse of the proposed green propellant pair does not present an advantage if compared to traditional hypergolic propellants, it offers a competitive advantage in terms of density-specific impulse, with the highest value achieved in the ignition delay time optimization, where the density-specific impulse of the system reached  $267.5 \text{ gs/cm}^3$ . Furthermore, the addition of n-butanol effectively reduced fuel viscosity, enhanced density-specific impulse, increased specific impulse, and improved ignition delay time response with 90% hydrogen peroxide compared to pure monoethanolamine formulations for a specific chamber and nozzle configuration. These findings highlight the potential of this green propellant system to enhance performance and efficiency in aerospace applications.

## 1. Introduction

Hydrazine and its derivatives have been used as fuel for auxiliary engines or ballistic missiles for many years by countries like China, Russia, and USA. In particular, bipropellant pairs like Nitrogen tetroxide (NTO or  $\text{N}_2\text{O}_4$ )/ Monomethylhydrazine (MMH) or NTO/ Unsymmetrical Dimethylhydrazine (UDMH) are used in attitude or re-acton control systems (for orbit change, reentry, or space rendezvous), planetary missions, terminal maneuvers, post boost control systems, and in missiles [1]. Retired projects like the Titan missile used nitrogen tetroxide and Aerozine 50 (a fuel mixture of 50% UDMH and 50% hydrazine) or the Space Shuttle Orbiter, which used MMH and NTO.

More recent ones, like the 10-N and 200-N bipropellant thrusters from Ariane Group, use MMH and UDMH with NTO pure or mixed with oxides of nitrogen (MON-1 and MON-3).d

However, these hypergolic combinations are extremely toxic, corrosive and carcinogenic [2], leading to more expensive projects due to additional steps in the operation, more safety provisions, additional design features, longer check-out procedures, and often more trained personnel [1]. In this context, alternative green propellants are being developed to overcome the adverse effects of highly volatile and toxic conventional propellants. With green propellants, safety handling requirements can possibly be less expensive, which offers a high cost-reducing potential [3].

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Recent research has focused on  $\text{H}_2\text{O}_2$  as an alternative for  $\text{N}_2\text{O}_4$ , a household product and environmentally friendly oxidizer that has been used successfully in bipropellant and monopropellant applications due to its low vapor pressure and non-toxic gas production upon combustion [2].

Studies conducted at Purdue University have shown that using hydrogen peroxide with manganese oxide ( $\text{Mn}_3\text{O}_4$ ) in a methanol fuel solution demonstrated rapid and reproducible ignition under single- and multiple-pulse firings in a bipropellant thruster, achieving an ignition delay time of 15 ms with 98% hydrogen peroxide [4].

Various studies have compared the performance of hydrogen peroxide for bipropellant thrusters, often in comparison with traditional hydrazine-based hypergolic pairs. According to P. Joy et al. [5], the  $\text{H}_2\text{O}_2$ /RP-1 propellant combination provides a better specific impulse for a selected mixture ratio and nozzle geometry and is classified as a green propulsion system, offering significant cost benefits over existing storable stages of the Indian Space Research Organization. The study indicated a payload gain of 103 kg compared to traditional bipropellants used for stage launchers.

Furthermore, the assumption that high-purity hydrogen peroxide can be safely used as an oxidizer for rocket motors has been confirmed by Okninski et al. [6]. Scaling effects in the transfer of technology from subscale laboratory research to in-flight operations were conducted, with key milestones in the viability of 98% hydrogen peroxide rocket motors established at The Lukaszewicz Research Network—Institute of Aviation in Warsaw, Poland, and through the ILR-33 AMBER, a suborbital Polish rocket [6].

Hydrogen peroxide reacts exothermically in the presence of a catalyst (often transition metal salts), which is typically suspended or dissolved in the fuel for bipropellant engines. A catalytically promoted (CP) fuel, like Block 0 (a mixture of 78 wt% methanol and 22 wt% manganese (II) acetate tetrahydrate), ignited with a delay of 13.32 ms upon contact with 98 wt%  $\text{H}_2\text{O}_2$ . However, catalytically promoted fuel includes salts of transition metals, including copper, iron, and chromium, that have an impact on the rocket engine's combustion performance parameters. Therefore, optimizing catalyst loading is required for any propellant combination in order to reduce ignition delay in CP fuels without sacrificing engine performance [7].

On the other hand, monoethanolamine (MEA) has been approved as a low-toxicity fuel compared to hydrazine. Various fuels were investigated as hypergolic pairings for  $\text{H}_2\text{O}_2$  90%, and MEA catalyzed by  $\text{CuCl}_2$  produced the best results. In addition, the hypergolicity of MEA with  $\text{H}_2\text{O}_2$  85% and 10%  $\text{CuCl}_2$  as catalyst was investigated. In this case, the ignition delay time was approximately 30 ms [8]. Studies made by Florczuk and Rarata [9] showed that the only use of MEA with  $\text{CuCl}_2$  as catalyst varies between 30 ms and 200 ms depending on the %wt of catalyst in the fuel.

In addition, studies on droplet intermixing, heat transfer, droplet swelling, and flame propagation for a hypergolic binary collision of 92%  $\text{H}_2\text{O}_2$  and monoethanolamine (MEA)-sodium borohydride ( $\text{NABH}_4$ ) droplets have been experimentally conducted by Zhang et al. [10]. Authors like Caffiero et al. [11] investigated the theoretical performance and ignition delay time of various MEA-based fuels, evaluating the influence of additive type, additive content, hydrogen peroxide grade, and the use of ethanol as a co-solvent. They achieved ignition delay times of 22.4 and 27.3 ms with 87.5%  $\text{H}_2\text{O}_2$  for MEA plus 10 wt% of copper(II) nitrate trihydrate and MEA plus 5 wt% of copper(II) chloride anhydrous, respectively. Increasing the oxidant purity to 98% for the same fuels proposed by the author resulted in ignition delay times of 16.3 and 18.5 ms.

Recent studies, like the one conducted by Maschio et al. [8] suggests that the inclusion of an alcohol to the mixture of MEA and catalyst could decrease the ignition delay time (IDT), increase the theoretical specific impulse ( $I_{sp_{th}}$ ) and that a well-grounded method of optimization (CCD method) can be used also for green propellant composition enhancement. N-butanol is a hydrophobic alcohol that can be produced

from glucose, and its inclusion in a fuel solution has a few advantages when compared with other alcohols. It has a higher volumetric energy density and lower vapor pressure than ethanol, and it can decrease the viscosity of the liquid [12].

According to Lapuerta et al. [13], adding n-butanol to a biodiesel solution lowers the freezing point of the mixture. This finding suggests that including n-butanol in an MEA solution may also reduce its freezing point compared to pure MEA. Since both MEA and n-butanol are fully miscible, and MEA has a relatively high freezing point (10.5 °C) compared to n-butanol (−90 °C) [14], mixing them in any proportion generally yields a solution with a freezing point lower than that of pure MEA alone. This effect, known as freezing point depression, is common in solutions where the presence of a solute (in this case, n-butanol) disrupts the crystalline structure of the solvent (MEA), making it more difficult for the solvent to solidify [15].

The primary goal of this research is to develop a green, homogeneous and hypergolic fuel blend that is less toxic than the traditional propellants pairs and safe for the environment. Hence, this study has three objectives. The first stage is selecting a catalyst that promotes the exothermic decomposition of 90% hydrogen peroxide and can combine with monoethanolamine and n-butanol homogeneously. Next, the IDT and the  $I_{sp_{th}}$  of the hypergolic reaction will be used, independently, as a unique response parameter in a central composite design (CCD) to optimize the composition of the n-butanol-monoethanolamine-catalyst mixture. The ultimate goal is to select the oxidizer/fuel ratio (O/F) for the propellant compositions in order to achieve the maximum  $I_{sp_{th}}$  computed by CEA NASA software for chemical equilibrium at 0.5 MPa combustion chamber pressure, 0.096 MPa exit pressure and nozzle expansion area ratio of 1.55.

## 2. Methodology

### 2.1. Drop test setup

A common technique for determining the IDT of hypergolic propellants and confirming their hypergolicity is the drop test. When using the drop test method, IDT values are measured differently from when performing the rocket injector test (i.e., impinging test). The varying oxidizer-to-fuel ratios and the impact of propellant mixing may be the cause of this difference. However, the drop test is a quick and straightforward method for determining a hypergolic propellant's IDT [16]. Fig. 1 depicts a schematic of the drop test configuration. During the experiment, a high-speed camera, PCO Dimax S1, was used, and the recordings were made at 3000 frames per second (fps), which introduced a systematic error of  $\pm 0.4$  ms.

The steps for the drop test are described below and Fig. 2 represents a flowchart for the drop test performed in this study.

1. Preparation of 5 mL homogeneous fuel solution (mixture of n-butanol, MEA and catalyst) in a becker.
2. Positioning the becker in the middle of the protection chamber in front of the camera.
3. With a burette, a small volume of  $\text{H}_2\text{O}_2$  (90%) is taken, and the burette is positioned with the valve closed 20 cm above the fuel level of the becker.
4. Preparation of the camera for the footage take. All the AC lights are turned off to minimize the noise, and the DC battery lights are turned on.
5. Drop one single droplet of  $\text{H}_2\text{O}_2$  (90%) on the fuel solution.
6. After recording the footage, the video is analyzed to determine the IDT. The number of frames between the contact of the  $\text{H}_2\text{O}_2$  (90%) droplet and the fuel pool until the reaction/combustion of the mixture are counted.

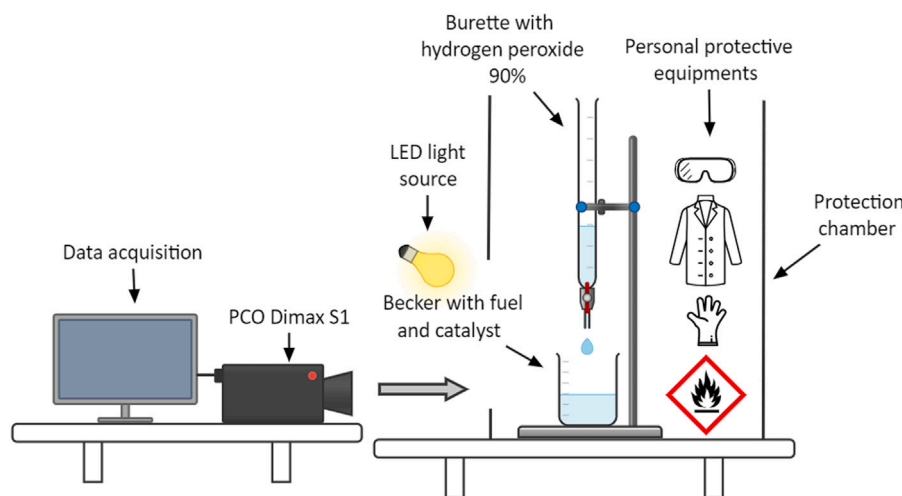


Fig. 1. Schematic of the drop test setup.

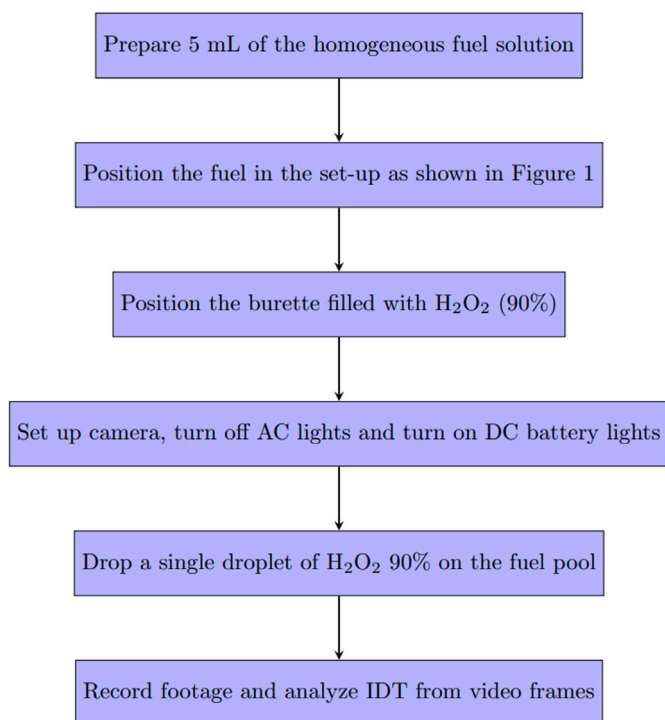


Fig. 2. Drop test process flowchart.

## 2.2. Catalyst candidates

The catalyst plays a fundamental role in the hypergolic reaction, as it promotes the exothermic decomposition of hydrogen peroxide ( $\text{H}_2\text{O}_2(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g}) + \text{O}_2$ ), generating enough heat to initiate the ignition process. Therefore, tests were initially carried out using  $\text{H}_2\text{O}_2$  (90%) and monoethanolamine promoted with different catalysts, in order to verify the activity of these compounds in the hypergolicity of the reaction. All catalyst candidates were used in preliminary tests, resulting in fuels containing 60% MEA, 30% n-butanol and 10% catalyst.

Due to the variety of bonding and coordination numbers it has, copper exhibits a wide range of chemical and spectroscopic characteristics. Hence, several cupric salt complex geometries with monoethanolamine are feasible [8]. Therefore, two copper compounds are proposed as

catalyst candidates, copper nitrate trihydrate ( $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ ) and copper chloride, both previously confirmed as hypergolic additives for monoethanolamine and  $\text{H}_2\text{O}_2$  (90%) [17].

Sodium borohydride ( $\text{NaBH}_4$ ), among all alkaline borohydrides, is widely used as a catalyst in hypergolic propellants due to its reaction with hydrogen peroxide in high purity (> 90%) [18]. Because it yields a homogeneous solution with monoethanolamine,  $\text{NaBH}_4$  was chosen as a candidate for catalyst.

Manganese and iodide compounds are used as oxidizing agents due to their redox properties in analytical methods for the quantitative determination of hydrogen peroxide [19], therefore, manganese nitrate, manganese acetate, manganese sulfate, potassium iodide and cuprous iodide are chosen as candidates. In addition to these previously mentioned compounds, lead nitrate and cobalt acetate are also tested as catalyst candidates.

All candidates proposed in this section were hypergolic tested and evaluated through a drop test set up. The selection criterion of the catalyst was the IDT because using the same weight percentage of different catalysts for the calculation of the  $Isp_{th}$  in the CEA NASA software does not cause a variance of the result due to the non-participation of the catalyst in the chemical reaction.

## 2.3. IDT and $Isp$ optimization

### 2.3.1. Experimental region delimitation

Once selected the catalyst that produced the lowest IDT, exploratory tests were carried out, seeking to delimit an experimental region that covered different fuel formulations. Two conditions were important to restrict these boundaries. The first condition was the necessity of a homogeneous solution, establishing the maximum amount of solid catalyst that can solubilize in a mixture of n-butanol and MEA, and the second condition required the solution to show a hypergolic response with  $\text{H}_2\text{O}_2$  (90%) during a drop test.

### 2.3.2. Design of experiments

Following the selection of the experimental region, a number of formulations were evaluated with the objective of achieving the IDT curve response and the  $Isp_{th}$  curve response and the minimum and the maximum of those curves, respectively. In order to achieve this goal, a CCD for a mixture of MEA, butanol and catalyst was performed according to the methodology applied by Maschio et al. [8].

Usually, a  $2^k$  factorial design leads to numerous treatment combinations, where the number of replicates is restricted. In order to avoid this problem and be able to fit a second-order model to our problem, it was decided to use a CCD method with 3 center points.

**Table 1**  
Catalyst candidates and type of reaction with H<sub>2</sub>O<sub>2</sub> 90%.

| Exp | Catalyst candidate        | Reaction             | Ignition delay time (ms) |
|-----|---------------------------|----------------------|--------------------------|
| 1   | Copper nitrate trihydrate | Hypergolic           | 20.4 ± 0.4               |
| 2   | Copper chloride           |                      | 23.4 ± 0.4               |
| 3   | Sodium borohydride        |                      | 24.7 ± 0.4               |
| 4   | Manganese nitrate         | Fast decomposition   | –                        |
| 5   | Manganese acetate         |                      | –                        |
| 6   | Lead nitrate              |                      | –                        |
| 7   | Cobalt acetate            | Medium decomposition | –                        |
| 8   | Manganese sulfate         |                      | –                        |
| 9   | Potassium iodide          | Slow decomposition   | –                        |
| 10  | Cuprous iodide            |                      | –                        |

The initial problem involving monoethanolamine, catalyst and n-butanol would have been a three-level factorial problem. Nonetheless, because the total sum of the substances must equal 1, we can simplify and use only 2 factors to obtain a quadratic model for the ignition delay response [20] (Eq. (1)).

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{ii} x_i^2 + \sum_{i < j} \sum_{i < j} \beta_{ij} x_i x_j + \varepsilon \quad (1)$$

Hence, the n-butanol load ( $\xi_B$ , with  $x_B$  as its coded level) and the catalyst content in the fuel ( $\xi_C$ , with  $x_C$  as its coded level) are the independent variables (factors) taken into account in the study. The relationship between the real and coded values is given by Eq. (2).

$$x = (\xi - \xi_0)/\Delta \quad (2)$$

Where 0 is the actual value located in the middle of the experimental range, 2A, which is made up of the initially chosen maximum and minimum experimental values. These restrictions are represented, respectively, by  $-1$  and  $1$  in the coded values. By rotating the starting experimental zone by 45 degrees, the final range of research is increased to include  $-\sqrt{2}$  to  $\sqrt{2}$  for both variables. As a result, each variable has five levels of operation ( $-\sqrt{2}$ ,  $-1$ ,  $0$ ,  $1$ ,  $\sqrt{2}$ ) [8].

### 2.3.3. Analysis of variance (ANOVA)

The determined quadratic model for the IDT and  $Isp_{th}$  response can be evaluated through the analysis of variance (ANOVA), where it is verified if the model fits the data collected and how is the behavior of the standardize residuals and the lack of fit of the model [20].

### 2.4. O/F and specific impulse

When talking about a propulsion system used in mini-satellites, where thrust generation and minimum pulse are important, trade-offs are necessary in order to achieve an effective solution for each mission proposed [21]. The first part of this work was related to optimizing the delay ignition response and the  $Isp_{th}$  of the fuel composition. The second part had the goal of selecting the oxidant-fuel ratio (O/F) for each fuel optimization (IDT and  $Isp_{th}$ ) that delivers the highest  $Isp_{th}$ . In the  $Isp_{th}$  optimization, this process just delivered the O/F ratio because the  $Isp_{th}$  is already the highest obtained throughout the CCD method. This analysis was made using the NASA CEA software.

## 3. Results and discussions

### 3.1. Catalyst selection

The ignition results of the reaction between the developed fuel and hydrogen peroxide (wt. 90%) are shown in Table 1.

The drop test of the hypergolic fuels with H<sub>2</sub>O<sub>2</sub> oxidizer is documented in Fig. 3. During the ignition phase, the oxidizer droplet's

**Table 2**  
Points in the ternary area chosen.

| Points | wt.% copper nitrate | wt.% n-butanol | wt.% MEA |
|--------|---------------------|----------------|----------|
| A      | 13                  | 0              | 87       |
| B      | 9                   | 31             | 60       |
| C      | 7.5                 | 42.5           | 50       |
| D      | 6                   | 54             | 40       |
| E      | 5                   | 45             | 50       |
| F      | 4                   | 36             | 60       |
| G      | 4                   | 0              | 96       |

contact time with the fuel pool was taken to be zero. Overall, the liquid-phase reaction, gaseous phase reaction and ignition can be distinguished from one another. The fuel pool and oxidizer drop combine to form an initiating structure, the liquid-phase reaction. Following this initial phase, the gaseous phase reaction appears, as shown in Fig. 3 by the visible gas generation and the following ignition process.

Given that it demonstrated the quickest hypergolic reaction (Table 1), copper nitrate trihydrate, called for simplicity as copper nitrate in this work, was chosen as a promoter for the decomposition of hydrogen peroxide (90%).

### 3.2. Experimental region selected for the CCD method

The experimental region inside the ternary plot is represented in Fig. 4, and it was a key result of this study, resulting in a limited zone for further optimization works with the proposed hypergolic pair. Table 2 shows the mass percentage values of each component that determine the boundaries of the experimental region.

The points A and G are, respectively, the maximum and minimum catalyst in wt.% for pure MEA, where the maximum of the mixture indicates the higher amount of catalyst in wt.% that the mixture or pure MEA can hold in a homogeneous solution and the minimum represents the least wt.% value of copper nitrate in the mixture or pure MEA that creates a hypergolic mixture with 90% hydrogen peroxide.

The amount of copper nitrate able to solubilize in a mixture of n-butanol and MEA is not constant due to the insolubility of copper nitrate in n-butanol, therefore, there is a reduction of the maximum as we walk towards higher percentages of n-butanol (segment ABCD in Fig. 4).

At the same time, formulations with wt.% of MEA between 40% and 60% resulted in an increment in the minimum catalyst wt.% of the mixture (segment DEF). In addition, mixtures of fuel where MEA percentage is inferior to 40% did not show hypergolic ignition with 90% hydrogen peroxide.

### 3.3. Experimental design via CCD method

The project of the experiment with the real values is presented in Fig. 5 and it follows the CCD project methodology [20].

The selection of the range of copper nitrate in the mixture, previously found in the experimental region, from 4 wt% to 13 wt%, served as the basis to define the experimental design (Fig. 5). Using these values, it was calculated the middle of this range, a value equal to 8.5 wt% (center point, 5, in Fig. 5).

For the n-butanol, the selected range for the center point (represented by the points 7 and 9 in Fig. 5) was from 0 to 36 wt% due to an initial estimation of where the optimized point would be, in case of failure of the optimization a new range for n-butanol should be settled. Therefore, the 45-degree-rotated square formed of points 6, 7, 8, and 9 was drawn first with a center, point 5, followed by the non-rotated square composed by the points 1, 2, 3 and 4.

The 11 experimental results using coded variables are displayed in Table 3. To reduce the impact of other variables, all trials were randomized and conducted on the same day.



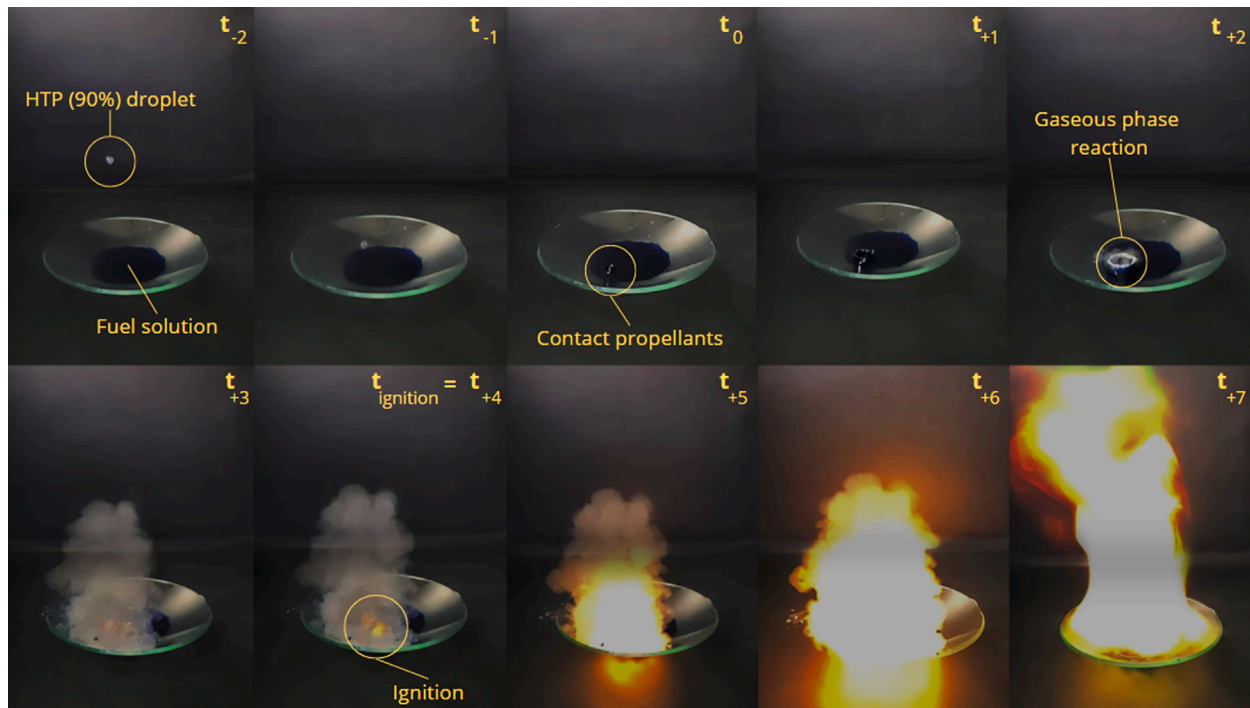


Fig. 3. Ignition images of copper nitrate as catalyst candidate.

Table 3  
Experimental results.

| Experimental level | Catalyst level ( $x_C$ ) | n-butanol level ( $x_B$ ) | Weight % Catalyst ( $\xi_C$ ) | Weight % n-butanol ( $\xi_B$ ) | IDT (ms)   | $Isp_{th}$ (s) |
|--------------------|--------------------------|---------------------------|-------------------------------|--------------------------------|------------|----------------|
| 1                  | -1                       | -1                        | 5.3                           | 5.27                           | 45.9 ± 0.4 | 167.2          |
| 2                  | -1                       | 1                         | 5.3                           | 30.7                           | 23.4 ± 0.4 | 170.8          |
| 3                  | 1                        | -1                        | 11.7                          | 5.27                           | 37.0 ± 0.4 | 166.3          |
| 4                  | 1                        | 1                         | 11.7                          | 30.7                           | 22.1 ± 0.4 | 170.3          |
| 5                  | 0                        | 0                         | 8.5                           | 18                             | 24.4 ± 0.4 | 168.8          |
| 5                  | 0                        | 0                         | 8.5                           | 18                             | 26.7 ± 0.4 | 168.8          |
| 5                  | 0                        | 0                         | 8.5                           | 18                             | 23.1 ± 0.4 | 168.8          |
| 6                  | -1.41                    | 0                         | 4                             | 18                             | 33.0 ± 0.4 | 169.3          |
| 7                  | 0                        | 1.41                      | 8.5                           | 36                             | 21.1 ± 0.4 | 171.3          |
| 8                  | 1.41                     | 0                         | 13                            | 18                             | 29.0 ± 0.4 | 168.3          |
| 9                  | 0                        | -1.41                     | 8.5                           | 0                              | 45.2 ± 0.4 | 165.8          |

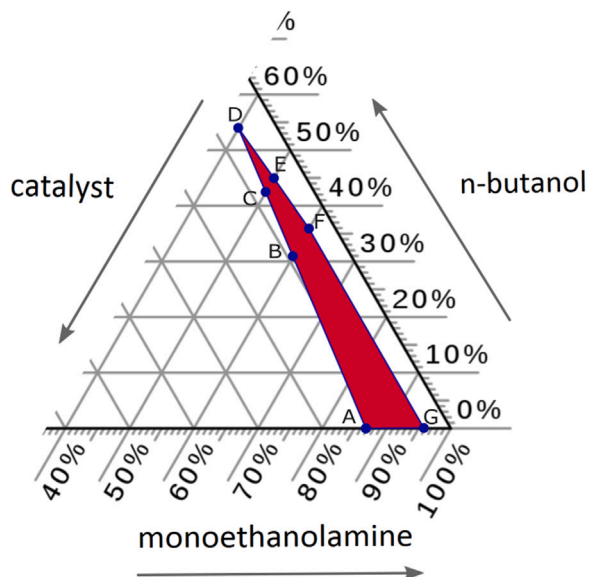


Fig. 4. Experimental region chosen for the CCD.

The determined quadratic model for the IDT response and the  $Isp_{th}$  are presented in Eqs. (3) and (4), respectively, with the coded variables. Eqs. (5) and (6) provide the same models expressed in real variables, measured in weight percent.

$$IDT = 24.33 - 1.83x_C - 8.88x_B + 1.75x_Cx_B + 3.28x_C^2 + 4.28x_B^2 \quad (3)$$

$$Isp_{th} = 168.78 - 0.34x_C + 1.92x_B - 0.10x_Cx_B - 0.0036x_C^2 - 0.13x_B^2 \quad (4)$$

$$IDT = 41.8 - 0.57\xi_C - 0.7\xi_B + 0.04(\xi_C - 8.5)(\xi_B - 18) + 0.32(\xi_C - 8.5)^2 + 0.03(\xi_B - 18)^2 \quad (5)$$

$$Isp_{th} = 166.65 - 0.11\xi_C + 0.15\xi_B - 0.0025(\xi_C - 8.5)(\xi_B - 18) - 0.00035(\xi_C - 8.5)^2 - 0.0008(\xi_B - 18)^2 \quad (6)$$

### 3.4. ANOVA

The significance of the quadratic regression can be evaluated through the analysis of variance (ANOVA). But, before applying the

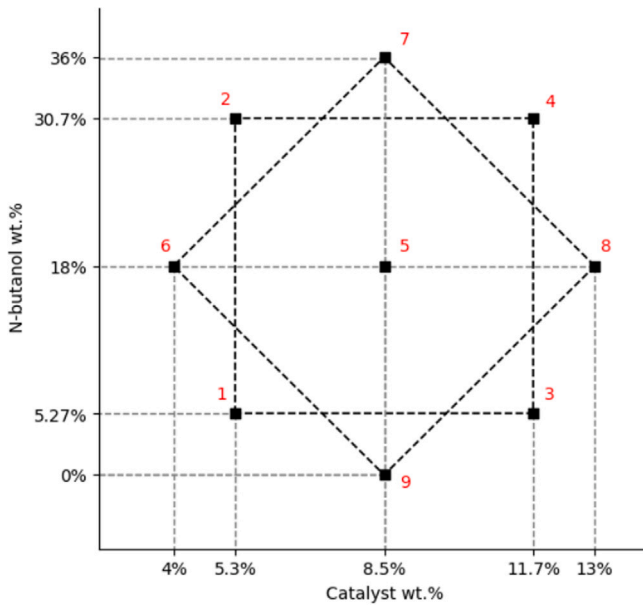


Fig. 5. Experimental design.

Table 4

Analysis of variance for the IDT fitted model proposed.

| Source      | DF | $SS_{adj}$ | $MS_{adj}$ | $F_{calc}$ | $F_{crit}$ |
|-------------|----|------------|------------|------------|------------|
| Regression  | 5  | 804.57     | 160.91     | 75.76      | 4.74       |
| $x_C$       | 1  | 31.35      | 31.35      | 14.76      | 6.61       |
| $x_B$       | 1  | 636.59     | 636.59     | 299.70     | 6.61       |
| $x_C * x_C$ | 1  | 55.32      | 55.32      | 26.05      | 6.61       |
| $x_B * x_B$ | 1  | 99.83      | 99.83      | 47.00      | 6.61       |
| $x_C * x_B$ | 1  | 14.40      | 14.40      | 6.78       | 6.61       |
| Residue     | 5  | 10.62      | 2.12       |            |            |
| Lack of fit | 3  | 3.87       | 1.29       | 0.38       | 19.16      |
| Error       | 2  | 6.75       | 3.37       |            |            |
| Total       | 10 | 815.19     |            |            |            |

ANOVA method a Levene's test is necessary to guarantee equal variances across samples, called homogeneity of variance. This test was performed through the statistical software R.

The p-value for Levene's test for both factors, catalyst and n-butanol, gives us values of 0.66 and 0.32, respectively, for the IDT optimization and 0.56 and 0.46 for the  $Isp_{th}$ . Those values are greater than the p-value for 95% confidence level (0.05), therefore, the null hypothesis that the samples to be compared come from a population with the same variance is accepted.

Tables 4 and 5 show the ANOVA performed for both optimizations. Per factor evaluated, a  $F_{calc}$  was calculated, being the ratio between the values of the quadratic regression ( $MS_{adj}$ ) and the quadratic residue ( $MS_{res}$ ). The comparison of  $F_{calc}$  to the critical F-value ( $F_{crit}$ ), tabulated for the 95% confidence level, shows that  $F_{calc} > F_{crit}$  for all factors and their interactions of second order, except for the second order of the catalyst percentage ( $x_C * x_C$ ) for the  $Isp_{th}$  optimization (Table 5), indicating that the  $x_C^2$  term on Eq. (6) is not relevant for the response and could be eliminated from the Equation. Hence, the quadratic models for the IDT and  $Isp_{th}$  response, presented in Eqs. (5) and (6), are statistically relevant.

The residual sum of squares left by the model can be decomposed into two parts: one caused by random errors, and the other due to the lack of fit of the model. The model's adequacy can be evaluated using the second part. The other component cannot be altered. The more adequate the model, the smaller the  $F_{calc}$  for the lack of fit. Since  $F_{lof} < F_{crit}$ , the IDT model does not show a lack of fit (as indicated in the Table 4). In the case of Table 5 the lack of fit is not possible

Table 5

Analysis of variance for the  $Isp_{th}$  fitted model proposed.

| Source      | DF | $SS_{adj}$ | $MS_{adj}$ | $F_{calc}$ | $F_{crit}$ |
|-------------|----|------------|------------|------------|------------|
| Regression  | 5  | 30.56      | 6.11       | 10190.46   | 4.74       |
| $x_C$       | 1  | 0.94       | 0.94       | 1564.37    | 6.61       |
| $x_B$       | 1  | 29.48      | 29.48      | 49147.25   | 6.61       |
| $x_C * x_C$ | 1  | 7.38e-05   | 7.38e-05   | 0.12       | 6.61       |
| $x_B * x_B$ | 1  | 0.09       | 0.09       | 156.04     | 6.61       |
| $x_C * x_B$ | 1  | 0.04       | 0.04       | 72.81      | 6.61       |
| Residue     | 5  | 0.003      | 0.0006     |            |            |
| Lack of fit | 3  | 0.003      | 0.001      | Inf        | 19.16      |
| Error       | 2  | 0          | 0          |            |            |
| Total       | 10 | 30.561     |            |            |            |

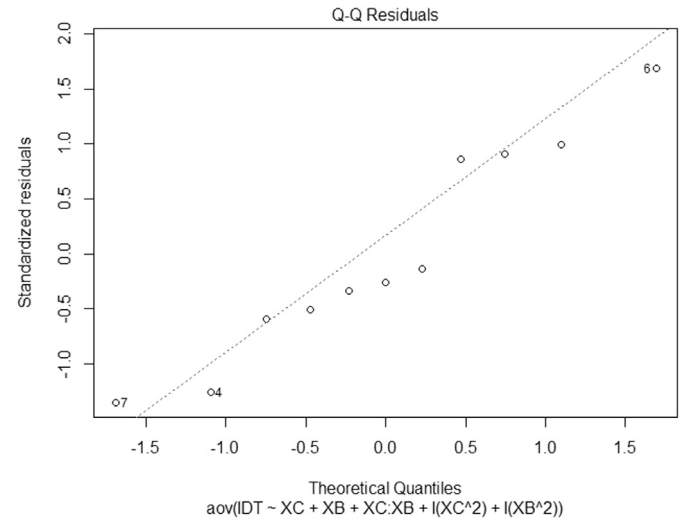


Fig. 6. Residual plot for the IDT ANOVA.

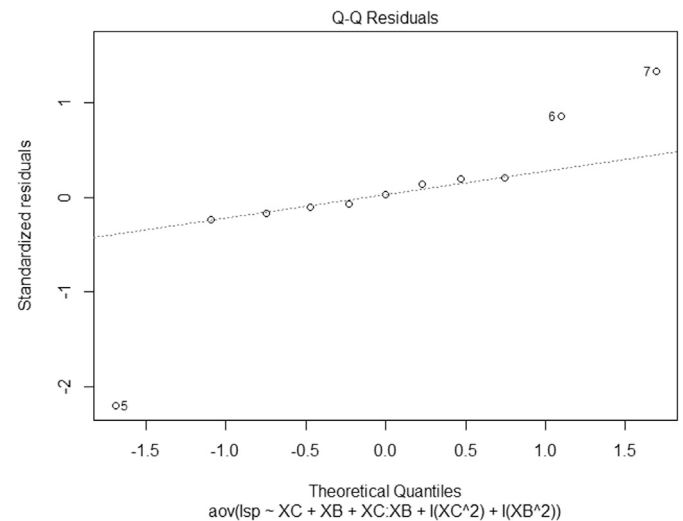


Fig. 7. Residual plot for the  $Isp_{th}$  ANOVA.

to evaluate properly since the center point, point 5 on Fig. 5, does not present a variance in the  $Isp_{th}$  (Table 3). R software was used to plot normal graphics for the residuals as an addition to the homoscedasticity analysis (Figs. 6 and 7). Therefore, the normal probability plots of the residuals suggest that the error terms are normally distributed for both optimizations.

Figs. 8 and 9 represent the model as response contour plots for IDT and  $Isp_{th}$ , respectively. All graphics were generated using Python.

**Table 6**  
Propellant composition.

| Propellant              | Composition (wt.%)                                                                 | Measured density (g/cm <sup>3</sup> )<br>at 298.15 K |
|-------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|
| IDT Optimization        | 60 MEA + 31.5 n-butanol + 8.5 Cu(NO <sub>3</sub> ) <sub>2</sub> ·3H <sub>2</sub> O | 0.904                                                |
| $Isp_{th}$ Optimization | 60 MEA + 36 n-butanol + 4 Cu(NO <sub>3</sub> ) <sub>2</sub> ·3H <sub>2</sub> O     | –                                                    |
| Oxidant                 | 90 H <sub>2</sub> O <sub>2</sub> + 10 H <sub>2</sub> O                             | 1.39                                                 |

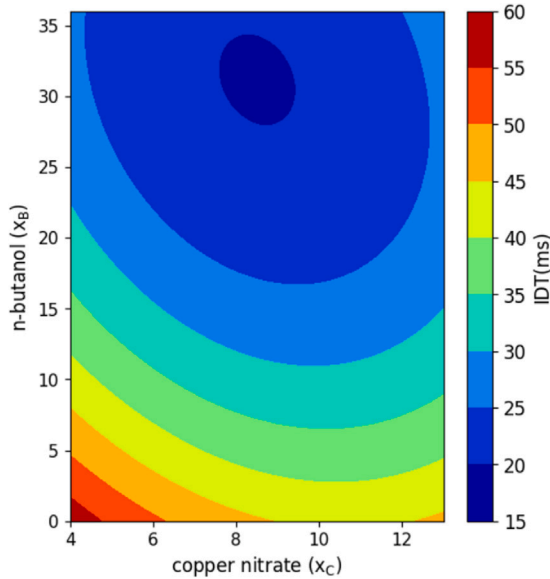


Fig. 8. IDT response contour plot.

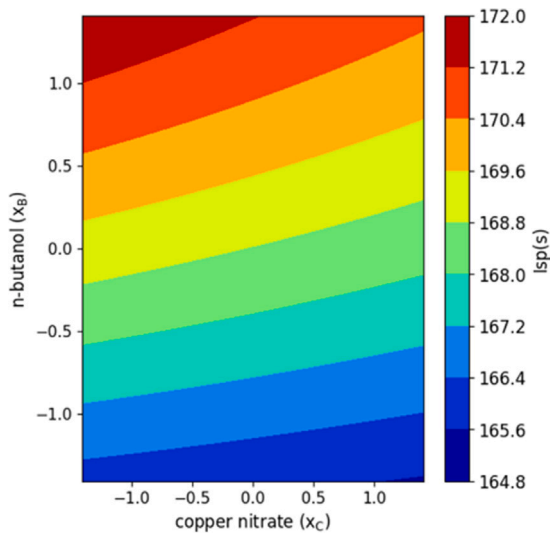


Fig. 9.  $Isp_{th}$  response contour plot.

The systems generated by the partial first derivatives of the fitted models (Eqs. (5) and (6)) can be used to determine the coordinates of the points of minimum for the IDT and maximum for the  $Isp_{th}$ . The system is solved when both of these equations equal zero. Hence,  $x_C = -0.004294$  and  $x_B = 1.059072$  are the points for the minimum IDT and  $x_C = -1.4$  and  $x_B = 1.4$  are the points for the maximum  $Isp_{th}$ . The best fit condition is shown in Table 6. Finally, the models predict a minimum IDT of  $20.0 \pm 0.4$  ms when the coded optimal values of  $x_C$  and  $x_B$  are substituted in Eq. (5) and a maximum  $Isp_{th}$  of 171.6 s when the coded optimal values are substituted in Eq. (6).

To verify the IDT model, a drop test with 90% hydrogen peroxide was used with the optimized fuel composition, and the result was a mean IDT of 21.5 ms with a standard deviation of  $\pm 1.30$  ms and a systematic error of  $\pm 0.4$ . It is crucial to draw attention to the reduction achieved by including n-butanol in the fuel composition. A test using monoethanolamine containing only 8.5% copper nitrate and 90% H<sub>2</sub>O<sub>2</sub> was conducted, and the ignition delay was  $45.2 \pm 0.4$  ms, as shown in Table 3.

### 3.5. O/F selection

Four fuel combinations were evaluated using the NASA CEA software for chemical equilibrium, utilizing 90% H<sub>2</sub>O<sub>2</sub> as oxidant: pure MEA with 8.5% of copper nitrate, the optimized fuel called “Ethanol”, containing 61% MEA, 30.1% ethanol and 8.9% copper nitrate proposed by Maschio et al. [22], the IDT optimized fuel composition developed in this paper using 31.5% n-butanol, 60% MEA and 8.5% copper nitrate, called “IDT Optimization” and the  $Isp_{th}$  optimized fuel composition using 36% n-butanol, 60% MEA and 4% copper nitrate, called “ $Isp_{th}$  Optimization”.

Fig. 10 shows the  $Isp_{th}$  curves for these 4 propellants. For both, pure MEA and “Ethanol” the optimum point is close to 3.5 O/F ratio, but when using n-butanol as an additive for MEA we have a shift of this O/F ratio to 4 (for the IDT Optimization) and to 4.4 (for the  $Isp_{th}$  Optimization), allowing not only a higher theoretical specific impulse but also increasing the total density of the propellant due to the higher density value of 90% hydrogen peroxide, therefore reducing the tank size of the propulsion system and the propellant consumption. Thus, an O/F ratio of 4 was selected for the IDT optimization and 4.4 for the  $Isp_{th}$  optimization.

Additionally, a  $\rho Isp_{th}$  (specific impulse density) comparison based on Maschio et al. [22] results is shown in Table 7. It considered an isentropic, one-dimensional flow in equilibrium condition during expansion, with stoichiometric conditions for all pairs except those specified with the O/F ratio, a chamber pressure of 5 bar, external pressure of 960 mbar, and an expansion rate of 1.55.

From the initial  $Isp_{th}$  assessment in Table 7, it can be concluded that the addition of n-butanol to the pure MEA solution increased the specific impulse of the propellant pair using H<sub>2</sub>O<sub>2</sub> (90%) as the oxidant. However, the  $Isp_{th}$  reached by the proposed hypergolic green pair is not superior to those commonly used in liquid rocket engines, such as MMH/NTO or N<sub>2</sub>H<sub>4</sub>/NTO.

Nevertheless, when the density of the reactants is taken into account, it is possible to deduce that in terms of achieving the same  $Isp_{th}$ , both optimized propellant pairs present an advantage in terms of energy content per unit volume of the propellant for a specific chamber and nozzle configuration. This increases its value when evaluating the overall mass requirements of a mission, especially when considering the volume constraints of the propellant tanks for in-space applications.

## 4. Conclusion

In conclusion, this study provides valuable insights into the optimization of a hypergolic green propellant pair based on n-butanol, monoethanolamine (MEA), and 90% hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>). Among 10 catalyst candidates, copper nitrate trihydrate demonstrated the most promising performance, igniting rapidly with MEA and n-butanol when paired with 90% H<sub>2</sub>O<sub>2</sub> as the oxidizer.

**Table 7**

Density of specific impulse and specific impulse for different propellants for a chamber pressure of 5 bar, external pressure of 960 mbar, and an expansion rate of 1.55.

Source: Adaptation from [22].

| Oxidant                             | Fuel                                                                                | O/F  | $Isp_{th}$ (s) | $\rho Isp_{th}$ (gs/cm <sup>3</sup> ) |
|-------------------------------------|-------------------------------------------------------------------------------------|------|----------------|---------------------------------------|
| LOX                                 | Ethanol                                                                             | 1.5  | 190.6          | 190.8                                 |
| LOX                                 | RP-1                                                                                | 2.1  | 198.0          | 205.1                                 |
| NTO                                 | MMH                                                                                 | 1.7  | 194.2          | 240.4                                 |
| NTO                                 | N <sub>2</sub> H <sub>4</sub>                                                       | 1.0  | 198.4          | 245.3                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | RP-1                                                                                | 7    | 179.3          | 221.6                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | Ethanol                                                                             | 4.5  | 176.2          | 226.4                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | N <sub>2</sub> H <sub>4</sub>                                                       | 1.8  | 188.8          | 243.9                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | 91.5% MEA + 8.5% Cu(NO <sub>3</sub> ) <sub>2</sub> ·3H <sub>2</sub> O               | 3.5  | 165.8          | 224.1                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | 30.1% Ethanol + 61% MEA + 8.9% Cu(NO <sub>3</sub> ) <sub>2</sub> ·3H <sub>2</sub> O | 3.86 | 170.0          | 223.6                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | IDT Optimization Fuel                                                               | 4    | 170.2          | 267.5                                 |
| H <sub>2</sub> O <sub>2</sub> (90%) | $Isp_{th}$ Optimization Fuel                                                        | 4.4  | 171.6          | 263.9                                 |

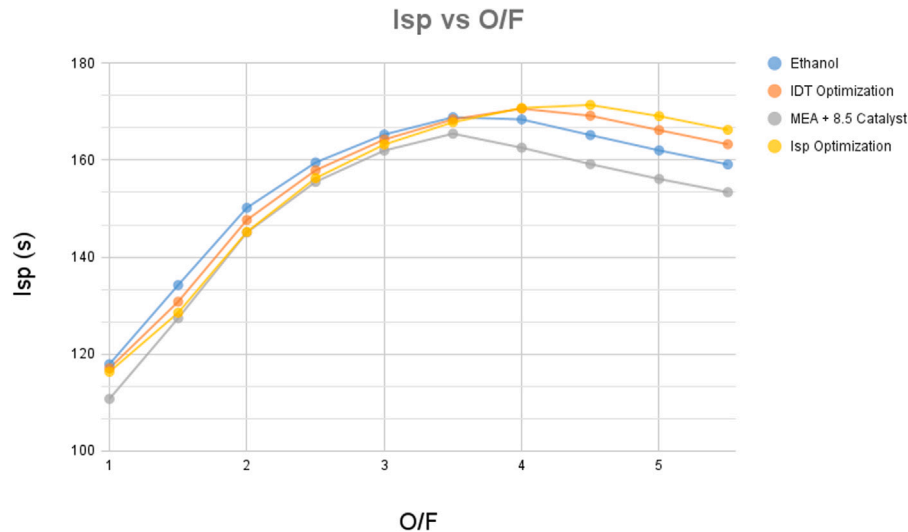


Fig. 10.  $Isp_{th}$  plot vs O/F ratio using 90% hydrogen peroxide.

The application of central composite design (CCD) methodology enabled the development of quadratic models for both specific impulse ( $Isp_{th}$ ) and ignition delay time (IDT), allowing for precise optimization. ANOVA tests confirmed the significance of these models in predicting propellant performance.

Optimization efforts identified the composition of 60% MEA, 31.5% n-butanol, and 8.5% copper nitrate as yielding the minimum IDT, validated through drop test setups with an IDT of 21.5 ms with a standard deviation of  $\pm 1.30$  ms and a systematic error of  $\pm 0.4$ . For IDT optimization, an oxidizer-to-fuel (O/F) ratio of 4 was found optimal, achieving a maximum  $Isp_{th}$  of 170.2 s.

Further refinement to maximize  $Isp_{th}$  resulted in a composition of 60% MEA, 36% n-butanol, and 4% copper nitrate at an O/F ratio of 4.4, achieving an  $Isp_{th}$  of 171.6 s with an IDT of  $26 \pm 0.4$  ms as predicted by the quadratic IDT model.

The addition of n-butanol to pure MEA increased the maximum  $Isp_{th}$  from 165.8 s to 171.6 s, resulting in an increment of 3.5%. A higher increment if compared with the addition of ethanol studied by Maschio et al. [8]. The maximum  $Isp_{th}$  reached by the green propellant pair proposed did not present an advantage if compared with the traditional hypergolic propellants  $Isp_{th}$ . However, it does present a competitive advantage in terms of  $\rho Isp_{th}$  for a specific chamber and nozzle configuration, with the highest value for the IDT optimization, where the  $\rho Isp_{th}$  of the system was equal to 267.5 gs/cm<sup>3</sup>.

Furthermore, the inclusion of n-butanol effectively reduced fuel viscosity, enhanced  $\rho Isp_{th}$ , increased specific impulse, and improved IDT response with 90% H<sub>2</sub>O<sub>2</sub> compared to pure MEA formulations. These findings underscore the potential of the studied green propellant system for enhancing performance and efficiency in aerospace applications.

## CRediT authorship contribution statement

**Paul C. Acosta Mendoza:** Writing – original draft, Formal analysis. **Rene F.B. Gonçalves:** Writing – review & editing, Supervision. **Leonardo Henrique Gouvêa:** Writing – review & editing, Supervision. **Luís Gustavo Ferroni Pereira:** Writing – review & editing, Visualization, Methodology, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## References

- [1] G.P. Sutton, O. Biblarz, *Rocket Propulsion Elements*, John Wiley & Sons, 2016.
- [2] V.K. Bhosale, J. Jeong, J. Choi, D.G. Churchill, Y. Lee, S. Kwon, Additive-promoted hypergolic ignition of ionic liquid with hydrogen peroxide, *Combust. Flame* 214 (2020) 426–436.
- [3] F. Lauck, J. Balkenhohl, M. Negri, D. Freudenmann, S. Schlechtriem, Green bipropellant development—a study on the hypergolicity of imidazole thiocyanate ionic liquids with hydrogen peroxide in an automated drop test setup, *Combust. Flame* 226 (2021) 87–97.



- [4] B.L. Austin, S. Heister, W. Anderson, Characterization of pintle engine performance for nontoxic hypergolic bipropellants, *J. Propuls. Power* 21 (4) (2005) 627–635.
- [5] G. P. Joy, et al., Development of green bi-propellant stage, in: *Proceedings of the International Conference on Aerospace & Mechanical Engineering (ICAME 21)*, 2022.
- [6] A. Okninski, P. Surmacz, B. Bartkowiak, T. Mayer, K. Sobczak, M. Pakosz, D. Kaniewski, J. Matyszewski, G. Rarata, P. Wolanski, Development of green storable hybrid rocket propulsion technology using 98% hydrogen peroxide as oxidizer, *Aerospace* 8 (9) (2021) 234.
- [7] J. John, P. Nandagopalan, S.W. Baek, S.J. Cho, Hypergolic ignition delay studies of solidified ethanol fuel with hydrogen peroxide for hybrid rockets, *Combust. Flame* 212 (2020) 205–215.
- [8] L.J. Maschio, L.G.F. Pereira, W.M. Meyer, R.I. Marques, R. Vieira, A DOE study on the hypergolicity of hydrogen peroxide with a rocket liquid fuel based on monoethanolamine and ethanol, *Int. J. Energetic Mater. Chem. Propuls.* 17 (2) (2018).
- [9] W. Florczuk, G. Rarata, Assessment of various fuel additives for reliable hypergolic ignition with 98%+ HTP, in: *66th International Astronautical Congress, International Astronautical Federation Jerusalem*, 2015.
- [10] D. Zhang, D. Yu, Y. Yuan, P. Zhang, X. Fan, Hypergolic ignition induced by head-on collision of bi-propellant droplets: Monoethanolamine-based fuel and hydrogen peroxide, *Fuel* 342 (2023) 127788.
- [11] L. Caffiero, S. Carlotti, F. Maggi, Monoethanolamine-based fuels hypergolic with rocket-grade hydrogen peroxide, *Aerospace* 11 (4) (2024) 309.
- [12] S. Sakai, D. Rothamer, Auto-ignition propensity of ethanol-, isobutanol-, and 2-methyl-3-buten-2-ol-gasoline blends under premixed prevaporized conditions in a spark-ignition engine, *Combust. Flame* 251 (2023) 112685.
- [13] M. Lapuerta, J. Rodríguez-Fernández, D. Fernández-Rodríguez, R. Patiño-Camino, Cold flow and filterability properties of n-butanol and ethanol blends with diesel and biodiesel fuels, *Fuel* 224 (2018) 552–559.
- [14] J.L. Richard, *Hazardous Chemicals Desk Reference*, John Wiley & Sons, Incorporated, 2003.
- [15] P.W. Atkins, J. De Paula, J. Keeler, *Atkins' Physical Chemistry*, Oxford University Press, 2023.
- [16] K.-S. Kim, V.K. Bhosale, S. Kwon, Synergistic effect of a hybrid additive for hydrogen peroxide-based low toxicity hypergolic propellants, *Combust. Flame* 231 (2021) 111450.
- [17] B. Melof, M. Grubelich, Investigation of hypergolic fuels with hydrogen peroxide, in: *37th Joint Propulsion Conference and Exhibit*, 2001, p. 3837.
- [18] H. Kang, J. Won, S.W. Baek, S. Kwon, Autoignition and combustion characteristics of sodium borohydride-based non-toxic hypergolic fuel droplet at elevated temperatures, *Combust. Flame* 181 (2017) 149–156.
- [19] Y.W. Kang, M.-J. Cho, K.-Y. Hwang, Correction of hydrogen peroxide interference on standard chemical oxygen demand test, *Water Res.* 33 (5) (1999) 1247–1251.
- [20] H. Rushing, A. Karl, J. Wisnowski, *Design and Analysis of Experiments by Douglas Montgomery: A Supplement for using JMP*, SAS Institute, 2014.
- [21] J.R. Wertz, D.F. Everett, J.J. Puschell, *Space mission engineering: the new SMAD*, 2011.
- [22] L.J. Maschio, E.P. de Araújo, L.G.F. Pereira, L.H. Gouvêa, R. Vieira, Assessing the performance of a green liquid fuel hypergolic with hydrogen peroxide in a 50 N bipropellant thruster, *Int. J. Energetic Mater. Chem. Propuls.* 20 (1) (2021).